Joseph Bennett Zeru

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Education

Bachelor of Science, Computer Science and Mathematics May 2028
Rose-Hulman Institute of Technology Terre Haute, IN
Certificates: Quantum Science, Computing, & Engineering
Minors: Imaging, Music

Skills

Relevant Courses: Deep Learning, Machine Learning, Linear Algebra I, Differential Equations II, Data Structures & Algorithms, Intro to Proofs, Combinatorics, Intro to Probability with Applied Statistics, Intro to Web Programming, Intro to Digital Systems
Coding Languages: Java, Python, C++, C, JavaScript, AngularJS, Verilog, Assembly, MATLAB, Simulink
Frameworks/Libraries: FastAPI, Angular, PyTorch/TensorFlow
Developer Tools: Git, LaTeX, Quartus, Firebase

Project Experience:

Document Privacy Analyzer
Associated with BinaryHeart *September 2025 - March 2026*

- Developed a web app to look through legal documents for clauses that exploit a customer's data and user privacy
- Fine-tuned a pre-trained model using public Terms of Service documents to create risk score calculations
- Built a responsive Angular front-end to present the risk analysis and a simplified summary for the user

EditorTrees
CSSE230 – Data Structures & Algorithms *October 2025*

- Applied knowledge of Binary & AVL Trees and their properties to implement efficient traversal algorithms
- Improved overall efficiency by reimplementing previously developed O(N) methods into O(log N)
- Implemented recursive methods to improve code readability and improve the general algorithmic processes

Work Experience:

BinaryHeart Terre Haute, IN
Project Lead and Rose-Hulman Chapter Co-Founder *June 2025 – Present*

- Lead logistical projects on computer distribution across the greater Terre Haute area and Africa
- Develop technical projects that provide the public with easy-to-use automation tools
- Promote initiatives with executive members through advertising and marketing strategies across campus to promote the club and improve its growth

BMS Software Team Member Terre Haute, IN
BattChallenge RHIT – Vigo Voltage *September 2025 – January 2026*

- Utilized MATLAB and Simulink to code test condition optimizations for efficient battery systems in motor vehicles
- Developed software requirements to improve workflow by 30% of car battery subsystems such as plug-in charging
- Organized and cleaned BattLab cleanup to help improve workflow and efficiency across multiple teams

Leadership and Activities:

Vice President Terre Haute, IN
BinaryHeart @ RHIT, 501(c)(3) *September 2025 – Present*

- Taught students with little to no background in computer maintenance, how to repair out-of-life computers
- Donated over $3000 worth of refurbished computers globally from greater Terre Haute to universities in West Africa
- Improved overall connection between

Boba Cart Electrical Team Lead Terre Haute, IN
Engineers for Sustainable World *September 2025 – November 2025*

CSSE Mentor Terre Haute, IN
CSSE Mentor/Mentee Program *September 2025 – December 2025*

Executive Board Member Terre Haute, IN
RISE Entrepreneurship *December 2024 – February 2025*